

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – 3.3 VECTORS I: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why did the Kisumu ferry miss Mbita, and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind? Use what you learned in Lessons 1 to 8 to answer each part below. Write in full sentences, show every step of your working, and include units. Draw neat labelled diagrams on grid paper where asked. Remember the situation: a ferry leaves Kisumu port to travel due east to Mbita, 60 km away. The engine pushes it east at 20 km/h, while a Lake Victoria current pushes it south at 8 km/h.

Part 1: Scalars, vectors and why direction matters

(a) Name two quantities from the ferry story. Say which is a scalar and which is a vector, and explain how you decided. (b) Give one more example of a scalar and one more example of a vector from Kenyan life (for example Kericho, Mombasa or a matatu trip from Nairobi to Nakuru). (c) The captain said, 'My engine pointed at Mbita, so I should have arrived at Mbita.' Using the words scalar and vector, explain what the captain forgot.

(a) The engine speed of 20 km/h is a quantity with magnitude only if we ignore where it points, so on its own it is a scalar. The engine thrust pointing east at 20 km/h is a vector, and so is the current pushing south at 8 km/h, because each has magnitude and direction. The distance of 60 km is a scalar. The displacement from Kisumu to Mbita is a vector. I decide by asking: do I need a direction to describe it completely? (b) Scalar: the temperature in Kericho, e.g. 18 °C. Vector: the wind from Mombasa blowing at 15 km/h towards the north-west, or the displacement from Nairobi to Nakuru in a particular direction. (c) The captain treated the ferry's motion as only a speed (a scalar) pointing east. But the current was a second vector acting at the same time. Directions matter, so the east vector and the south vector combine into one resultant vector that points south of east, not due east. That is why the ferry missed Mbita.

Part 2: Writing and adding the vectors

(a) Write the engine thrust and the lake current as column vectors, using east as the positive horizontal direction and north as the positive vertical direction. (b) Write the position vector of Mbita relative

(a) Engine thrust = $(20, 0)$. Current = $(0, -8)$. (b) Mbita is 60 km east and 0 km north or south of Kisumu, so its position vector is $(60, 0)$. (c) Resultant = $(20, 0) + (0, -8) = (20, -8)$ km/h, adding the top entries and the bottom entries separately. (d) Triangle law: draw the engine arrow 20 units east, then draw the current arrow starting at the tip of the engine arrow, 8 units south (tip to tail). The resultant goes from the start of the first arrow to the tip of the second. Parallelogram law: draw both arrows from the same starting point, complete the parallelogram, and the diagonal from that point is the resultant. Both give the same arrow, $(20, -8)$. (e) The current acts southward, which is the negative vertical direction. The bottom entry of the resultant records this downward push, so the ferry's real path has a southward part, unlike the intended path $(60, 0)$, which has none.

to Kisumu port (the origin) for the captain's intended journey of 60 km east. (c) Add the engine and current column vectors to find the resultant vector. (d) Describe how you would show this addition using the triangle law, and say how the parallelogram law gives the same result. (e) Explain why the resultant vector has a negative bottom entry.	
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Part 3: How fast and in what direction did the ferry really go?

(a) Calculate the magnitude of the resultant vector. Show the formula you use and give your answer to 1 decimal place. (b) Calculate the angle between the resultant path and due east, and state it in words such as 'south of east'. (c) The ferry needs 3 hours to cover 60 km at its engine speed of 20 km/h. Calculate how far south the current pushes it in that time. (d) A classmate says: 'The ferry's speed over the lake is $20 + 8 = 28$ km/h.' Explain why this is wrong.	(a) The east and south components are perpendicular, so they form a right-angled triangle and I use Pythagoras: $R = \sqrt{(20^2 + 8^2)} = \sqrt{(400 + 64)} = \sqrt{464} \approx 21.5$ km/h. (b) $\tan \theta = 8/20$, so $\theta = \arctan(0.4) \approx 21.8^\circ$. The ferry travels about 21.8° south of east (a bearing of about 111.8°). (c) Drift south = $8 \text{ km/h} \times 3 \text{ h} = 24$ km. The ferry lands about 24 km south of the point it was aiming for. (d) Adding 20 and 8 would only be correct if both pushes were in the same direction. Here they are at right angles, so they are the two legs of a right-angled triangle and the resultant is the hypotenuse. Direction forces us to use squares and a square root, giving about 21.5 km/h, which is less than 28 km/h.
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Part 4: Correcting the course

(a) Suppose a ferry has landed at a wrong point L and needs to reach the intended point M. Explain in words how you find the correction vector from L to M using position vectors. (b) Explain what the negative of a vector is, and what you get when you add a vector to its negative. (c) The captain wants	(a) The correction vector from L to M is the position vector of M minus the position vector of L. I subtract the top entries and the bottom entries separately, which is the same as adding the negative of the position vector of L. (b) The negative of a vector has the same magnitude but the opposite direction. Adding a vector to its negative gives the zero vector, which means no net displacement, like a round trip. (c) The northward part of the engine velocity must equal the southward current of 8 km/h. If θ is the angle north of east, then $20 \sin \theta = 8$, so $\sin \theta = 0.4$ and $\theta \approx 23.6^\circ$. The captain should point the ferry about 23.6° north of east (a bearing of about 066.4°), not due east. The northward part of the engine then cancels the current's southward push, and the ferry's path is directed towards Mbita. The captain pre-compensates for the current instead of ignoring it.
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to keep the engine speed at 20 km/h but tilt the heading north of east so that the northward part of the engine velocity exactly cancels the southward current. Find the angle north of east for the heading, and say what this tells the captain to do differently from before. Give your answer to 1 decimal place.	
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Part 5: What if the current gets stronger? Answering the driving question	
(a) Suppose the current doubles. Write the new current as a column vector using scalar multiplication, and say whether it is parallel to the original current. How do you know? (b) Calculate the new resultant column vector and its magnitude when the engine is still (20, 0). (c) Use your results from Parts 1 to 4 to write a short paragraph (4 to 6 sentences) answering the driving question: why did the ferry miss Mbita, and how does vector mathematics help the captain always reach the right destination?	(a) New current = $2 \times (0, -8) = (0, -16)$. It is parallel to the original current because it is a scalar multiple of it ($k = 2$). Same direction, bigger magnitude. (b) Resultant = $(20, 0) + (0, -16) = (20, -16)$ km/h. Magnitude = $\sqrt{(20^2 + 16^2)} = \sqrt{(400 + 256)} = \sqrt{656} \approx 25.6$ km/h. The ferry would drift south twice as fast, so over 3 hours it would be pushed $16 \times 3 = 48$ km south, twice as far as before. (c) The ferry missed Mbita because the captain treated the journey as a speed pointing east, a scalar idea, but the engine thrust and the lake current are two vectors acting at the same time. Added together they make a resultant pointing about 21.8° south of east, so the ferry landed south of Mbita. Column vectors let us write each push precisely, vector addition combines them, and Pythagoras and trigonometry give the size and direction of the resultant. Vector subtraction and the right heading calculation show the captain how to correct the course, here about 23.6° north of east. Because direction is part of a vector, the captain can pre-compensate for any known current or wind and always reach the right destination.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scalar and vector reasoning	Correctly classifies quantities as scalars or vectors with clear reasons, and explains fully how treating velocity as a scalar caused the ferry to miss Mbita.	Classifies most quantities correctly and gives a reasonable explanation of why direction matters in the ferry story.	Confuses scalars and vectors or does not link direction to the ferry missing Mbita.

Vector representation and addition	Writes all column and position vectors with correct signs, adds them accurately, and explains both triangle and parallelogram laws.	Writes and adds the column vectors correctly with minor sign or explanation slips.	Makes frequent errors in column vector signs or addition, or cannot describe the addition laws.
Magnitude, direction and calculation	Uses Pythagoras and arctan accurately with units and sensible rounding (about 21.5 km/h, 21.8° south of east, 24 km drift, 23.6° correction), and explains why $20 + 8$ is wrong.	Calculates most values correctly with working shown, with small rounding or unit errors.	Cannot set up the right-angled triangle or calculate magnitude and angle correctly.
Correction and scalar multiplication	Explains subtraction of position vectors, the negative and zero vector, and parallel scalar multiples, and applies them correctly to the corrected heading and doubled current.	Applies most of these ideas correctly with some gaps in explanation.	Does not connect subtraction or scalar multiplication to correcting the ferry's course.
Communication and link to the driving question	Writes a clear, connected answer to the driving question using correct vocabulary, labelled diagrams and Kenyan context.	Answers the driving question with mostly correct vocabulary and some connection of ideas.	Answer is incomplete or lists facts without linking them to the ferry phenomenon.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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